

Supplementary Material: Early Black-Hole Growth Constraints from JWST

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11 September 2026

The main paper reports required average Eddington ratios for 224 objects in its Balmer-line sample and 234 objects in the expanded sample. This supplement illustrates the change in growth when accretion begins at $z = 3400$ instead of $z = 30$. The main paper gives the seed-mass compatibility grid and its methods in an appendix.

S1 Growth equations

We use $f_{\text{Edd}} = L_{\text{bol}}/L_{\text{Edd}}$, radiative efficiency ϵ , seed mass M_{seed} , and merger multiplier B_{merge} . The time available for growth is $\Delta t = t(z_{\text{obs}}) - t(z_{\text{seed}})$. At fixed efficiency, growth depends on the time-average $\overline{f_{\text{Edd}}}$:

$$M_{\text{BH}}(t_{\text{obs}}) = B_{\text{merge}} M_{\text{seed}} \exp \left[\frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}}} \right], \quad (\text{S1})$$

$$\log_{10} \left(\frac{M_{\text{seed, req}}}{M_{\odot}} \right) = \log_{10} \left(\frac{M_{\text{BH}}}{M_{\odot}} \right) - \log_{10} B_{\text{merge}} - \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \Delta t}{t_{\text{Edd}} \ln 10}. \quad (\text{S2})$$

$$t(z) = \frac{2}{3H_0\sqrt{\Omega_{\Lambda}}} \operatorname{asinh} \left[\sqrt{\frac{\Omega_{\Lambda}}{\Omega_m}} (1+z)^{-3/2} \right], \quad (\text{S3})$$

Here $t_{\text{Edd}} = 0.45$ Gyr. The age relation uses $H_0 = 67.3 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$ and $\Omega_{\Lambda} = 0.685$ and neglects radiation. The main calculations begin at $z_{\text{seed}} = 30$. The extension to $z = 3400$ below is a mathematical extrapolation of this age relation.

S2 Earlier growth onset and primordial seed interpretations

The growth equations take the initial mass and subsequent accretion history as inputs. The seed-formation mechanism is left unspecified. A primordial black hole (PBH) already present at $z = 30$ can therefore be represented by its effective mass at that epoch if it lies within the broad 10^2 – $10^6 M_{\odot}$ interval and follows the adopted subsequent growth prescription. This mass need not equal its primordial formation mass. This interval covers a specified range of effective masses; inferring a primordial origin requires a separate formation model. In [P. Dayal \(2024\)](#), $z \simeq 3400$ marks matter–radiation equality and the start of the modelled evolution of PBHs that formed earlier.

Moving the start of the same constant-rate, constant-efficiency history from $z = 30$ to 3400 adds $\delta t = t(30) - t(3400) = 0.09990$ Gyr in our matter-plus-Lambda extrapolation. At fixed seed mass and merger

multiplier, the resulting shift is

$$\Delta \log_{10} M_{\text{BH}} = \frac{1 - \epsilon}{\epsilon} \frac{\overline{f_{\text{Edd}}} \delta t}{t_{\text{Edd}} \ln 10}. \quad (\text{S4})$$

For $\overline{f_{\text{Edd}}} = 1$, this is 0.868 dex (a factor of 7.37) at $\epsilon = 0.1$, or 1.589 dex (a factor of 38.85) at the ideal non-spinning thin-disc efficiency (J. M. Bardeen et al. 1972). The tracks shift upward by these amounts (Figure S1). At fixed observed mass and seed mass requiring positive accretion, the ratio of required average rates is $[t(z_{\text{obs}}) - t(30)]/[t(z_{\text{obs}}) - t(3400)]$; the earlier start lowers the requirement by 6.5, 13.1 and 22.9 per cent at $z_{\text{obs}} = 4, 7$ and 10.603, respectively.

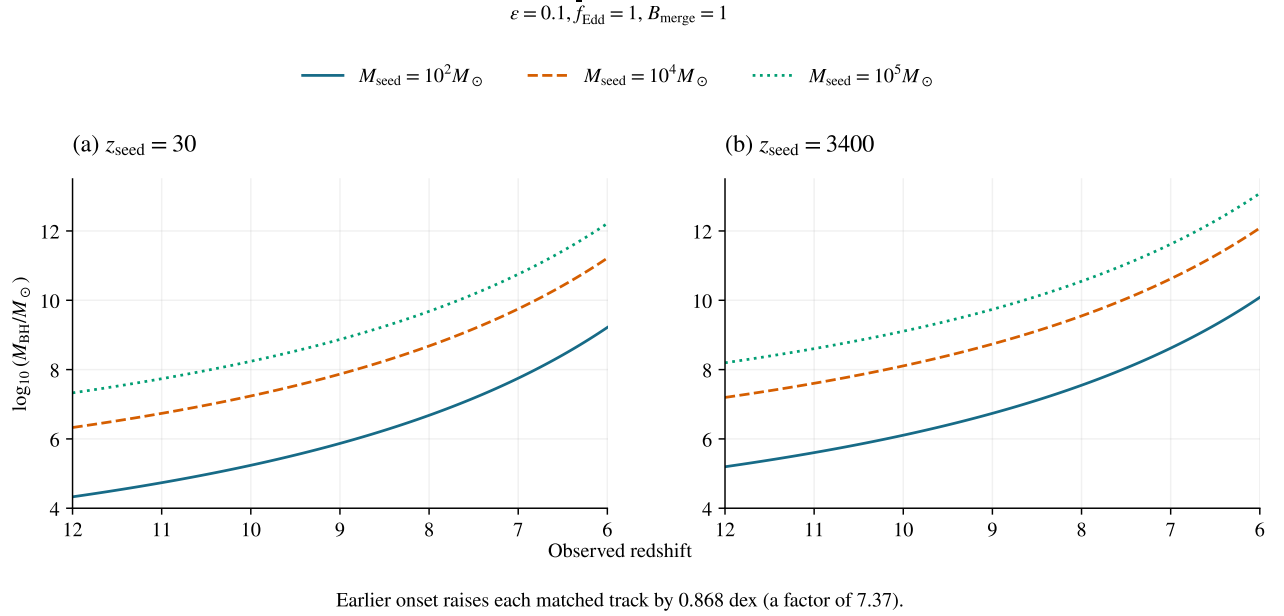


Figure S1: Growth from $z_{\text{seed}} = 30$ (a) and 3400 (b), with matched seed masses and the fixed assumptions shown. Earlier onset adds 99.90 Myr and shifts each track upward by 0.868 dex. Here z_{seed} denotes the assumed start of accretion. The $z = 3400$ comparison extrapolates the matter-plus-Lambda cosmology and assumes uninterrupted accretion. Radiation, gas supply, halo assembly and feedback are omitted.

These comparisons quantify the additional growth time. Applying them to PBHs requires an effective mass at the chosen starting epoch and a model for the subsequent gas supply and accretion history.

References

- Bardeen, J. M., Press, W. H., & Teukolsky, S. A. 1972, ApJ, 178, 347, doi: [10.1086/151796](https://doi.org/10.1086/151796)
- Dayal, P. 2024, A&A, 690, A182, doi: [10.1051/0004-6361/202451481](https://doi.org/10.1051/0004-6361/202451481)